

TIME TO LIVE

www.esempio.it

~~138.12.277.42~~

107.46.72.221

SERVER
DNS



ALTRI SERVER



equivalente ↴

```
andrea@aquinas:~  
$ nslookup www.mit.edu  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
www.mit.edu canonical name = www.mit.edu.edgekey.net.  
www.mit.edu.edgekey.net canonical name = e9566.dscb.akamaiedge.net.  
Name:   e9566.dscb.akamaiedge.net  
Address: 23.50.153.111  
Name:   e9566.dscb.akamaiedge.net  
Address: 2a02:26f0:2d:1182::255e  
Name:   e9566.dscb.akamaiedge.net  
Address: 2a02:26f0:2d:1191::255e
```

```
andrea@aquinas:~  
$ nslookup -type=A www.mit.edu  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
www.mit.edu canonical name = www.mit.edu.edgekey.net.  
www.mit.edu.edgekey.net canonical name = e9566.dscb.akamaiedge.net.  
Name:   e9566.dscb.akamaiedge.net  
Address: 23.50.153.111
```

Mx (Mail Exchange)



indirizzo IP del
mail server

```
andrea@aquinas:~  
$ nslookup -type=MX mit.edu  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
mit.edu mail exchanger = 100 mit-edu.mail.protection.outlook.com.  
  
Authoritative answers can be found from:
```

```
andrea@aquinas:~  
$ nslookup -type=A mit-edu.mail.protection.outlook.com  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
Name:   mit-edu.mail.protection.outlook.com  
Address: 52.101.11.3  
Name:   mit-edu.mail.protection.outlook.com  
Address: 52.101.8.44  
Name:   mit-edu.mail.protection.outlook.com  
Address: 52.101.42.14  
Name:   mit-edu.mail.protection.outlook.com  
Address: 52.101.40.1
```

DNS server con autorità



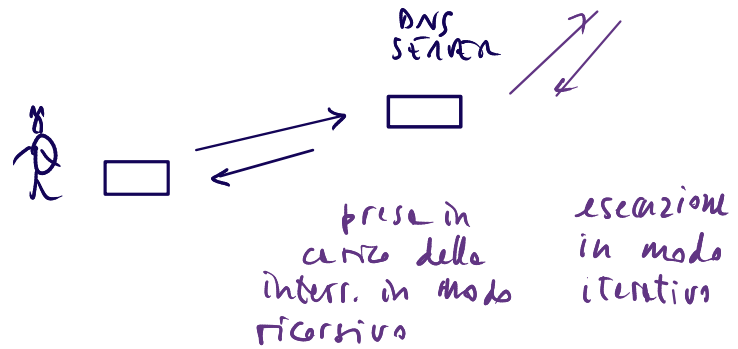
DNS server principale

```
andrea@aquinas:~  
$ nslookup -type=NS mit.edu  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
mit.edu nameserver = asia2.akam.net.  
mit.edu nameserver = asia1.akam.net.  
mit.edu nameserver = use2.akam.net.  
mit.edu nameserver = ns1-37.akam.net.  
mit.edu nameserver = usw2.akam.net.  
mit.edu nameserver = ns1-173.akam.net.  
mit.edu nameserver = use5.akam.net.  
mit.edu nameserver = eur5.akam.net.  
  
Authoritative answers can be found from:  
asia2.akam.net internet address = 95.101.36.64  
use5.akam.net internet address = 2.16.40.64  
usw2.akam.net internet address = 184.26.161.64  
asia1.akam.net internet address = 95.100.175.64  
use2.akam.net internet address = 96.7.49.64  
eur5.akam.net internet address = 23.74.25.64  
ns1-37.akam.net internet address = 193.108.91.37  
ns1-173.akam.net internet address = 193.108.91.173  
use5.akam.net has AAAA address 2600:1403:a::40
```

```
andrea@aquinas:~  
$ nslookup -type=SOA mit.edu  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
mit.edu  
origin = use2.akam.net  
mail addr = network-request.mit.edu  
serial = 21763  
refresh = 900  
retry = 900  
expire = 3600000  
minimum = 3600  
  
Authoritative answers can be found from:
```

ESEMPIO

```
andrea@aquinas:~  
$ nslookup -type=A hyoka.ofc.kyushu-u.ac.jp  
Server:      192.133.28.1  
Address:     192.133.28.1#53  
  
Non-authoritative answer:  
Name:   hyoka.ofc.kyushu-u.ac.jp  
Address: 133.5.40.197
```



RISOLUZIONE ITERATIVA

Ignoriamo il server DNS locale e risolviamo il nome iterativamente, come farebbe il DNS server in assenza di inform. nelle cache.

```
andrea@aquinas:~  
$ nslookup -type=A hyoka.ofc.kyushu-u.ac.jp c.root-servers.net  
Server:      c.root-servers.net  
Address:     192.33.4.12#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer
```

Il root server non sa nulla

```
andrea@aquinas:~  
$ nslookup -type=NS hyoka.ofc.kyushu-u.ac.jp c.root-servers.net  
Server:      c.root-servers.net  
Address:     192.33.4.12#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer  
  
Authoritative answers can be found from:  
jp      nameserver = b.dns.jp.  
jp      nameserver = d.dns.jp.  
jp      nameserver = f.dns.jp.  
jp      nameserver = g.dns.jp.  
jp      nameserver = c.dns.jp.  
jp      nameserver = e.dns.jp.  
jp      nameserver = h.dns.jp.  
jp      nameserver = a.dns.jp.  
h.dns.jp      internet address = 161.232.72.25  
g.dns.jp      internet address = 203.119.40.1  
f.dns.jp      internet address = 150.100.6.8  
e.dns.jp      internet address = 192.50.43.53  
d.dns.jp      internet address = 210.138.175.244  
c.dns.jp      internet address = 156.154.100.5  
b.dns.jp      internet address = 202.12.30.131  
a.dns.jp      internet address = 203.119.1.1  
h.dns.jp      has AAAA address 2a01:8840:1bc::25  
f.dns.jp      has AAAA address 2001:2f8:0:100::153  
e.dns.jp      has AAAA address 2001:200:c000::35  
d.dns.jp      has AAAA address 2001:240::53  
c.dns.jp      has AAAA address 2001:502:ad09::5  
b.dns.jp      has AAAA address 2001:dc2::1  
a.dns.jp      has AAAA address 2001:dc4::1
```

Il root server ci dice
gli indirizzi dei NS
per **jp**

```
andrea@aquinas:~  
$ nslookup -type=A hyoka.ofc.kyushu-u.ac.jp b.dns.jp  
Server:      b.dns.jp  
Address:     202.12.30.131#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer
```

Il server per jp
non sa nulla

```
andrea@aquinas:~  
$ nslookup -type=NS hyoka.ofc.kyushu-u.ac.jp b.dns.jp  
Server:      b.dns.jp  
Address:     202.12.30.131#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer  
  
Authoritative answers can be found from:  
kyushu-u.ac.jp  nameserver = dns-x.sinet.ad.jp.  
kyushu-u.ac.jp  nameserver = ns1.kyushu-u.ac.jp.  
kyushu-u.ac.jp  nameserver = ns2.kyushu-u.ac.jp.  
ns2.kyushu-u.ac.jp  internet address = 133.5.1.9  
ns1.kyushu-u.ac.jp  internet address = 133.5.6.3
```

Il server per jp conosce
gli indirizzi dei NS
per kyushu-u.ac.jp

```
andrea@aquinas:~  
$ nslookup -type=A hyoka.ofc.kyushu-u.ac.jp ns2.kyushu-u.ac.jp  
Server:      ns2.kyushu-u.ac.jp  
Address:     133.5.1.9#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer
```

(Come sopra)

```
andrea@aquinas:~  
$ nslookup -type=NS hyoka.ofc.kyushu-u.ac.jp ns2.kyushu-u.ac.jp  
Server:      ns2.kyushu-u.ac.jp  
Address:     133.5.1.9#53  
  
Non-authoritative answer:  
*** Can't find hyoka.ofc.kyushu-u.ac.jp: No answer  
  
Authoritative answers can be found from:  
ofc.kyushu-u.ac.jp  nameserver = ns.jimu.kyushu-u.ac.jp.  
ofc.kyushu-u.ac.jp  nameserver = dns-x.sinet.ad.jp.  
ns.jimu.kyushu-u.ac.jp  internet address = 133.5.40.1
```

(Come sopra)

```
andrea@aquinas:~  
$ nslookup -type=A hyoka.ofc.kyushu-u.ac.jp ns.jimu.kyushu-u.ac.jp  
Server:      ns.jimu.kyushu-u.ac.jp  
Address:     133.5.40.1#53  
  
Name:  hyoka.ofc.kyushu-u.ac.jp  
Address: 133.5.40.197
```

Finalmente abbiamo
interrogato il server DNS
con autorità sul nome
richiesto.